SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

SET 3 :

1. A manufacturing company plans to migrate its legacy enterprise applications and databases from an on-premises data center to the cloud. The organization wants to minimize downtime while ensuring application performance and security.

- Design a cloud migration strategy suitable for the organization.
- Explain how different migration approaches (Rehost, Refactor, Replatform, Repurchase, Retire, Retain) can be applied.
- Discuss methods to validate application performance after migration.
- Recommend strategies to reduce migration risks and downtime.

2. A smart city project deploys thousands of IoT sensors to monitor traffic, air quality, energy consumption, and public safety. The collected data must be processed and analyzed in real time for decision-making.

- Design a cloud-based architecture for IoT data collection and analytics.
- Explain how message brokers, stream processing, and cloud storage are integrated.
- Discuss methods for handling high-volume, high-velocity sensor data.
- Recommend techniques to ensure scalability, reliability, and security.

3. A global e-commerce company wants executives to monitor sales performance, customer behavior, and regional trends through interactive dashboards. The solution must support real-time analytics and historical reporting.

- Design a cloud-based business intelligence architecture.
- Explain how data visualization tools integrate with cloud data warehouses.
- Discuss techniques for optimizing dashboard performance with large datasets.
- Recommend security measures to protect business reports and sensitive data.

4. An online gaming company experiences fluctuating workloads throughout the day. During peak hours, users report high latency and reduced application performance.

- Design a cloud monitoring and performance optimization solution.
- Explain how cloud monitoring tools, metrics, logs, and tracing help identify performance bottlenecks.
- Discuss strategies for resource optimization and capacity planning.
- Recommend methods to improve application availability while controlling operational costs.

5. A financial institution stores petabytes of customer transaction data for regulatory compliance and business analytics. The organization must retain historical records while managing storage costs efficiently.

- Design a cloud-based data lifecycle management strategy.
- Explain how data classification, retention policies, and archival storage should be implemented.
- Discuss the importance of governance, auditing, and regulatory compliance in big data environments.
- Recommend techniques for balancing data accessibility, security, and cost optimization.

SET - 3 cloud computing

① cloud Migration Strategy for a manufacturing company

Cloud migration is process of moving Applications Databases, other IT Resources from an on-Premises data center to cloud platform such as AWS, Microsoft Azure or Google cloud

Cloud migration Strategy

Assessment

-Identify existing Applications, databases, dependencies, and hardware requirements

Planning

Select a suitable cloud Provider and migration method for each Application.

Backup

Create comple backups for Applications and databases

) c1c (29) Cloud - Based IOT Data collection and Analytics for smart city

A:

Application Migration

Move Application in Phases to reduce downtime

Testing

Verify Functionality, security, and Performance

Deployment

Redirect users to the cloud environment

Monitoring

Continiously monitor the migrated Applications

Applications:

Perform load testing to Simulate many users

conduct stress testing to identify system limits

Measure response time, CPU, storage.

Compare Application Performance.

Conduct User Acceptance testing

Test Database Queries and data Consistency.

(9) Cloud - Based IOT Data collection and Analytics for Smart City

A smart city uses IOT sensors to collect data about traffic, air, Quality, Energy consumption, and Public Safety. Cloud computing enables the real-time Processing

Cloud -Based Architecture:

IOT sensor: Collect traffic, Pollution and Energy data

IOT gateway: Connects Sensors to internet and Performs initial Processing

Message Broker: Receives and distributes sensor Messages

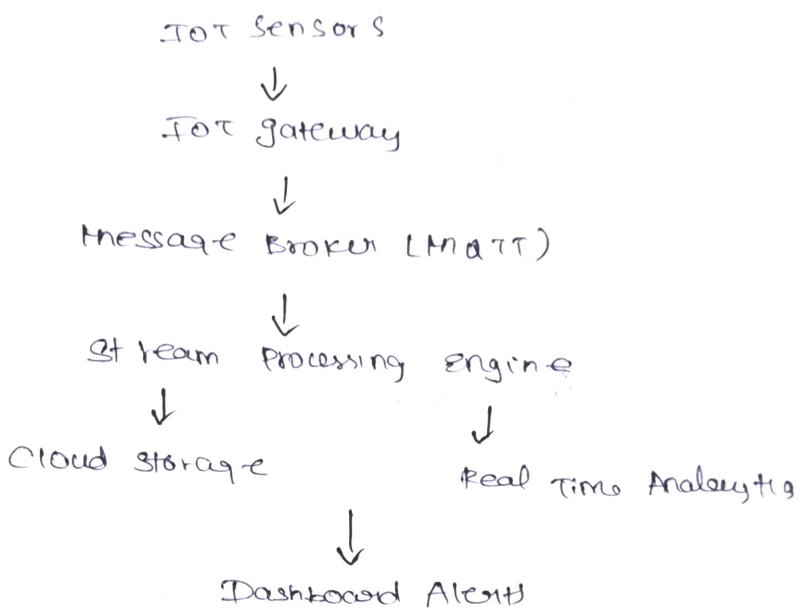
Stream Processing: Process incoming data in real-time

Cloud storage: Stores raw and Processed data

Analytics Engine: Identifies Patterns and Predicts Possible Problems

Dashboard Displays real time information to City Administrator

Architecture flow



Conclusion:

A cloud based IOT Architecture enables real-time monitoring, efficient data analysis and faster-decision making while providing scalability, reliability, security for smart cities.

5) Cloud-Based Business intelligence Architecture for E-commerce

Business Intelligence is the process of collecting Analyzing, Presenting business data to support decision-making. An e-commerce company can use cloud BI to monitor Sales, customer, Product, Performance, regional trends

Architecture flow

Sales, customer, Website and Inventory data

↓

Data ingestion (Batch/real time)

↓

ETL / ELT & Data cleaning

↓

Cloud Data Warehouse

↓

Analytics and Query Engine

↓

Interactive BI Dashboards & reports

Conclusion:

A cloud BI Architecture combines data integration, cloud warehouses, Analytics, & interactive Dashboards to Provide timely business insights.

④ Cloud Monitoring and Performance optimization online Gaming company

online gaming workloads fluctuate throughout the day. During peak hours, increases users can cause high latency, server overload, and poor gaming experience.

Cloud Monitoring optimization:

Game clients: send player request to the gaming Platform

load balancer: Distribute request across Available servers

Game servers: Process game play & user section

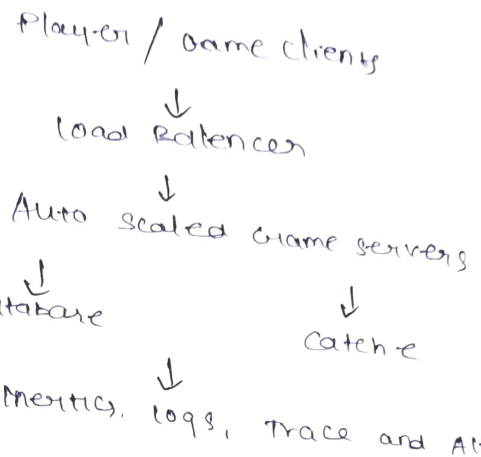
Alert System: notifies Administrators about failures & Performance issues

Game clients : send plc

gaming Platform

· Manage : Distribu

Architecture Flow



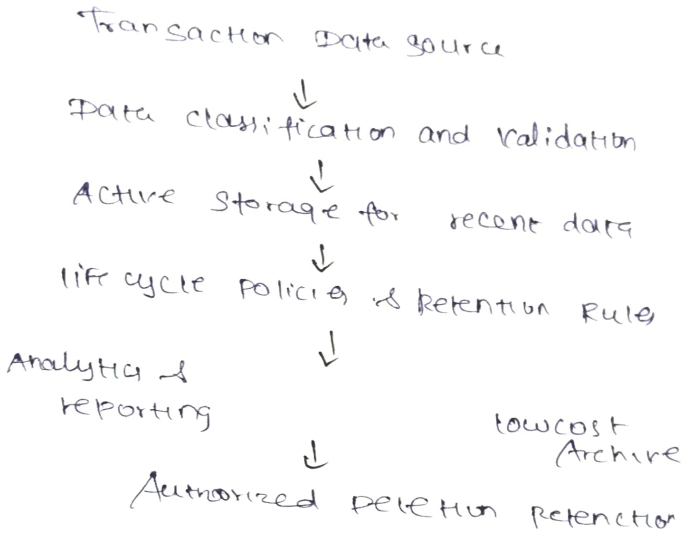
(5) Cloud - Based Data lifecycle management for financial institution.

A financial institution may store petabytes of customer transaction data for regulatory compliance, auditing, fraud detection, and business Analytics. Data lifecycle management organize data from creation to deletion while balancing accessibility, security, & storage cost.

load balancer :
Available servers

... ex: Prodcon game ...

Architecture flow



Importance :

Data governance
/Auditing
compliance
/Access control
Data integrity
Accountability

Conclusion..

A cloud-based data life cycle strategy reduces storage cost while preserving data availability, integrity, security, regulatory